

ASSIGNMENT - 5

Half -range fourier series

1. Obtain the Fourier expansion of x^2 as a cosine series in $(0, \pi)$.

$$\text{Ans: } f(x) = \frac{\pi^2}{3} - 4 \left[\cos x - \frac{1}{2^2} \cos 2x + \frac{1}{3^2} \cos 3x + \frac{1}{4^2} \cos 4x + \dots \right]$$

2. Obtain the half range cosine series for the function $f(x) = x$ in the interval $0 < x < 2$.

$$\text{Ans: } a_0 = \frac{-8}{\pi^2}, \quad a_n = \frac{4}{n^2 \pi^2} [(-1)^n - 1]$$

3. Obtain the half range sine series for the function $f(x) = t - t^2$ in the interval $0 < t < 2$.

$$\text{Ans: } f(x) = \frac{8}{\pi^3} \left[\frac{\sin \pi t}{1} - \frac{1}{3^3} \sin 3\pi t + \frac{1}{5^3} \sin 5\pi t + \dots \right]$$